Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find the surface area of a closed 2 by 3 by 5 cm box.

6. Five granola bars are shared equally among six children. How much does each get?

7. 60 ÷ 5

8. Find the distance from (1, -5) to (1, 4).

9. Reflect (-3, 4) across the y-axis.

10. Find the distance from (-4, 2) to (3, 2).

11. Write an inequality for x that means at least 9. Is the boundary included?

12. Write an inequality for x that means less than 5. Is the boundary included?

13. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

14. Write signed numbers for 6 meters below sea level and 5 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -6, 0, $\frac{-5}{2}$.

16. Find the opposite of −6, then its opposite. Explain why you return to the start.

17. Find |−6|, |6|, and |0|. Compare the first two.

18. Which is warmer, −6°C or −8°C? Which has greater absolute value?

CHALLENGE / Explain your plan and check the result

C1. A rectangle has vertices (−5, −2), (1, −2), (1, 3), (−5, 3). Find its side lengths, perimeter, and area. Reflect it across the x-axis. Give the reflected point corresponding to (−5, 3) and explain which measurements change.

C2. A package may weigh at most 8 kg. Its empty box weighs 1.5 kg. Two identical items are packed inside. Write an inequality for the weight w of each item. Test w = 3, w = 3.25, andw = 3.5 kg.